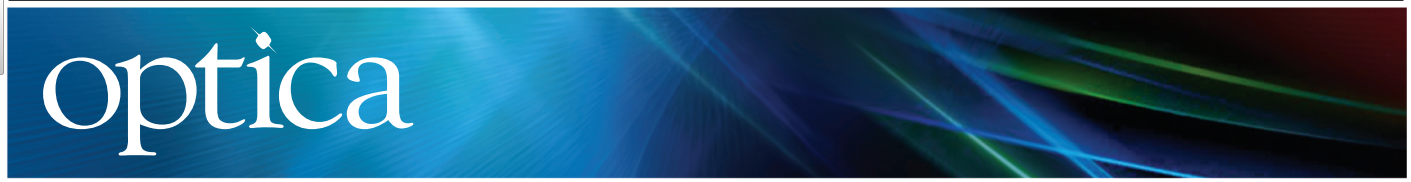
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Experimental simulation of time and frequency transfer via an optical satellite–ground link at 10^{-18} instability

Qi Shen,[1,2,3,†] Jian-Yu Guan,[1,2,3,†] Ting Zeng,[1,2,3] Qi-Ming Lu,[1,2,3] Liang Huang,[1,2,3] Yuan Cao,[1,2,3] Jiu-Peng Chen,[1,2,3] Tian-Qi Tao,[2,4] Jin-Cai Wu,[2,4] Lei Hou,[5] Sheng-Kai Liao,[1,2,3] Ji-Gang Ren,[1,2,3] Juan Yin,[1,2,3] Jian-Jun Jia,[2,4] Hai-Feng Jiang,[1,2,3,5] Cheng-Zhi Peng,[1,2,3,6] Qiang Zhang,[1,2,3,4,7] and Jian-Wei Pan[1,2,3,8]

[1]Shanghai Branch, National Laboratory for Physical Sciences at Microscale and Department of Modern Physics, University of Science and Technology of China, Shanghai 201315, China
[2]CAS Center for Excellence and Synergetic Innovation Center in Quantum Information and Quantum Physics, University of Science and Technology of China, Shanghai 201315, China
[3]Shanghai Research Center for Quantum Sciences, Shanghai 201315, China
[4]Key Laboratory of Space Active Opto-Electronic Technology, Shanghai Institute of Technical Physics, Chinese Academy of Sciences, Shanghai 200083, China
[5]Key Laboratory of Time and Frequency Primary Standards, National Time Service Center, Chinese Academy of Sciences, Xi'an 710600, China
[6]e-mail: pcz@ustc.edu.cn
[7]e-mail: qiangzh@ustc.edu.cn
[8]e-mail: pan@ustc.edu.cn

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Microwave-based satellite–ground links are used to transfer time and frequency in various applications such as metrology, navigation, positioning, and very long baseline interferometers. The existing approaches, however, cannot fully satisfy the requirements of these applications. In this study, we investigated the possibility of an optical-based satellite–ground link, where the transferred carriers are pulsed lasers, resulting in a link with a high time resolution and a large ambiguous range. First, we analyzed the parameters of satellites in different orbits and concluded that high-orbit links enable more stable time–frequency comparison or dissemination by taking advantage of the long duration, a large common view range, and the lower relativistic effects. Subsequently, we performed a 16 km free-space transfer experiment to simulate links in the loss, noise, and delay effects. The link exhibits an instability of 4×10^{-18} at 3,000 s and an approximately 10 fs time deviation with an average loss of 72 dB, corresponding to the loss of a satellite–ground link at geostationary earth orbit (GEO). Based on these results, we expect that the instability of the time–frequency transfer via a GEO link might reach 10^{-18} at 10,000 s. © 2021 Optical Society of America under the terms of the OSA Open Access Publishing Agreement

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1. INTRODUCTION

High-precision time–frequency dissemination between remote locations enables many applications, such as the redefinition of the second [1–4]; tests of general relativity [5,6] and fundamental quantum physics [7,8]; precision navigation [9,10]; and quantum communications [11]. Fiber-based optical time–frequency dissemination has demonstrated unprecedented instability over dedicated fiber networks [12–14]. However, applying fiber links in locations such as mountainous [15] and marine regions is difficult, whereas satellite-based optical time–frequency dissemination is a reasonable option for intercontinental dissemination. Traditional satellite-based links exhibit an optimum frequency instability of approximately 1×10^{-15} for a day, which is mainly limited by the resolution of the microwave carrier [16,17]. To improve the resolution, time transfer by a laser link (T2L2) with short laser pulse carriers has been explored. However, limitations due to the response speed of photoelectric detections makes it difficult to exceed the stability of the 10^{-17} level in a one-day period [18,19]. In fact, this performance is comparable to that of new-generation microwave links with multiple carriers for the atomic clock ensemble in space (ACES) project [20,21]. The planned European laser timing (ELT+) optical link for the I-SOC project is aimed at providing a frequency transfer uncertainty on the order of 10^{-18} for integration times longer than 10 days [21]. Currently, no satellite link can compare or transfer time–frequency signals generated by the best optical frequency standards with a stability of 10^{-18} level at 10,000 s [22].

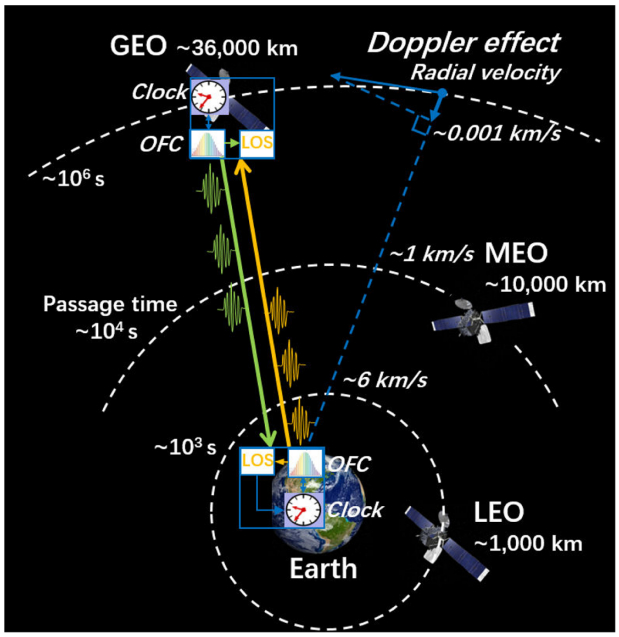


Fig. 1. Satellite–ground optical time–frequency dissemination diagram with different orbit types. The typical heights for LEO, MEO, and GEO are 1,000 km, 10,000 km, and 36,000 km, respectively. The downlink losses are 23.8 dB, 43.8 dB, and 54.9 dB, and the uplink losses are 40.0 dB, 60.0 dB, and 71.1 dB, respectively. The maximum radial velocities were approximately 6 km/s, 1 km/s, and 1 m/s, and the passage times are approximately 10^3 s, 10^4 s, and 10^6 s, respectively. The point-ahead angles are 45.9 μrad, 29.8 μrad, and 17.3 μrad (see Supplement 1).

A pioneering technique that combines optical frequency combs (OFCs) and linear optical sampling (LOS) [23] based on optical interference, which avoids any photoelectronic response limitation, can reach orders of 10^{-18} within 1,000 s on the ground with a static [24–26] or a movable platform [27]. In principle, stable satellite–ground links can be realized with this technique by using its high resolution and relatively large ambiguous range. An ambiguous range of a few nanoseconds, corresponding to the duration between the laser pulses of the OFC, enables accurate reconnection after suffering random signal fading due to the atmosphere. This is the most important advantage of the OFC-based link compared to a continuous-wave link, whose ambiguous range is usually below 5 fs. Figure 1 shows a possible simplified version of the satellite–ground optical time–frequency dissemination diagram. Two combs, phased-locked to optical atomic clocks, are located at the satellite and the ground station. Both sites can send and receive comb pulses through the telescopes. At each site, the received light then can be coupled into the single-mode fiber to beat with the local comb. Subsequently, the difference in the beat signals from the two sites can be calculated and used as the feedback signal to compare or synchronize the optical frequencies [24].

2. SATELLITE–GROUND LINK ANALYSIS

The fundamental question concerns the type of orbit to be used. Based on the experience pertaining to the Micius quantum satellite [28], we estimated the loss, Doppler speed, and time of passing territory of the link, as shown in Fig. 2. The downlink and uplink losses can be calculated using

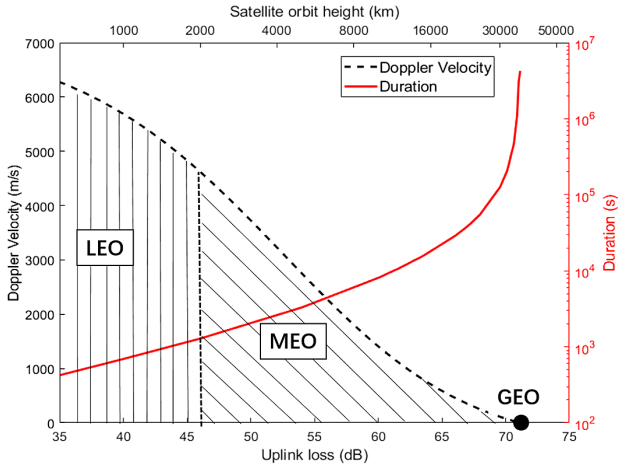


Fig. 2. Satellite–ground uplink loss, radial velocity, and duration of a single passage with different satellite orbit height. The satellite orbit was assumed to be circular, with a zero-degree inclination. The ground station was set in the equator line, and the field of view was 150° from the minimum elevation angle of 15°. The radial velocity was calculated with the satellite at an elevation angle of 15° from the horizon plane of the ground station. The different satellite orbits correspond to different areas in the figure. The orbit height of LEO is lower than 2,000 km, and the orbit height of MEO is between 2,000 km and 35,786 km, which is the height of the GEO. The pioneering works [24,26,27] show a link loss of about 50 dB and a Doppler velocity of up to 24 m/s, which can mimic the conditions expected for the MEO. The solid circle point represents this work with a 72 dB link loss, corresponding to the GEO conditions.

$$\eta_{\mathrm{down}} = \eta_{\mathrm{tele_s}}\left(\frac{D_g}{L\theta_{\mathrm{down}}}\right)^2 T_{\mathrm{atm}}\eta_{\mathrm{tele_g}}\eta_{\mathrm{sm_g}},$$

$$\eta_{\mathrm{up}} = \eta_{\mathrm{tele_g}}\left(\frac{D_s}{L\theta_{\mathrm{up}}}\right)^2 T_{\mathrm{atm}}\eta_{\mathrm{tele_s}}\eta_{\mathrm{sm_s}}, \quad (1)$$

where $\eta_{\mathrm{tele_s}}$ and $\eta_{\mathrm{tele_g}}$ are the telescope optical efficiency of the satellite and the ground, respectively; D_s and D_g are the apertures of the telescope; L is the satellite-to-ground distance; θ_{down} and θ_{up} represent the effective transmitter full-angle divergence for the downlink and the uplink, respectively; T_{atm} is the atmospheric transmittance, which is reduced by the air absorption and scattering of the propagating beam; and $\eta_{\mathrm{sm_s}}$ and $\eta_{\mathrm{sm_g}}$ represent the single-mode fiber coupling efficiency on the satellite and on the ground, respectively (see Supplement 1). The loss is calculated based on these parameters. Both telescopes at the satellite and the ground have a diameter of 1 m and an optical efficiency of 0.8, the transmitter full-angle divergence for the downlink and uplink is 4 μrad and 15 μrad, and the atmospheric transmittance is 0.7 for 1550 nm. The uplink loss is considerably higher than the downlink loss, which is mainly because of the different divergence angles and coupling efficiency of single-mode fibers from free space. In the uplink, atmospheric turbulence close to the ground strongly deteriorates the quality of the beam. Therefore, turbulence-induced distortion significantly increases the beam divergence angle for the uplink. In terms of the single-mode fiber coupling efficiency, for the downlink, the adaptive optics technology can be employed to optimize the coupling efficiency to more than 15% [29]; however, for the uplink, it is estimated to be only 5% because the adaptive optics is almost useless, and the single-mode coupling efficiency for a high-speed moving satellite is extremely sensitive to the optical axis consistency between the ground and the satellite telescope. As

shown in Fig. 2, the maximum link attenuation is estimated to be as large as 71 dB for the uplink of a geostationary earth orbit (GEO) with a 36,000 km satellite–ground distance.

For a low earth orbit (LEO), the typical height is approximately 1,000 km, and the transmission loss of the link is below 50 dB, which is comparable to that of a 4 km ground link [24]. Such a link has to overcome the strong Doppler effect [27] of more than 100 MHz/s frequency shift over 3.7 GHz, for a 1.5 μm OFC, corresponding to a radial acceleration of 0.1 km/s^2 and velocity of 5.6 km/s. The optical beat note of LEO would cross the blind range because the frequency shift is greater than the repetition rate of a common OFC. We believe that the multi-LEO design can solve this problem. However, the frequency uncertainty of the space clock will be a few 10^{-18}, limited by the orbit determination ability of a few centimeters and the relativistic effects. On the contrary, a short passing time and common view distance limit the time and frequency comparisons between intercontinental optical clocks. Therefore, the LEO link is more suitable to verify the fundamental physical rules such as the relativistic effect [30], rather than comparing to or transferring time and frequency of the best clocks.

Thus, medium earth orbit (MEO) and GEO are comparatively better options for the target orbit, where the instability due to relativistic effects would be well below 1×10^{-18} and the passing and common view times are substantially larger than those of the LEO, enabling the potential performance of optical atomic clocks and intercontinental comparison of ground clocks. Further, the Doppler effect is lower than that of the LEO link. A previous study [27] demonstrated a maximum radial velocity of 24 m/s, corresponding to a high earth orbit of 34,400 km. The distance between the back and forward transmission paths in air, corresponding to a point-ahead angle of 25 μrad, is below 0.25 m. The link instability due to noncoincidence of the optical paths was experimentally investigated through a few-kilometer link that mimicked the conditions expected for a ground-to-satellite link to be 10^{-18} at 1,000 s [31]. A theoretical analysis [32] reported an even better expectation that the asymmetry effect due to relative motion would be less than 2×10^{-17} at 1 s. Thus, for these orbits, the high loss, atmospheric noise, and long delay are the major challenges.

3. EXPERIMENTAL SETUP

To verify the possibility of precision time–frequency transmission via an MEO or GEO link, we constructed a free-space channel comprising a 16 km atmosphere to simulate a GEO link. As shown in Fig. 3(a), the 16 km atmosphere link was actually a folded 8 km atmosphere link, both of which are located at the Shanghai branch of the University of Science and Technology of China, two telescopes with a diameter of 300 mm and 280 mm, respectively, mounted on the roof of our laboratory building and a 500 mm diameter plane mirror placed at a room on the 17th floor located in the Huamu subdistrict of Pudong. A horizontal density air path of 16 km in a low-elevation noisy city is well beyond the effective aerosphere thickness of a GEO link, typically below 10 km [33]. The typical Fried parameter (r0) [34] of the ground 8 km atmosphere link is between 6 and 10 cm. In fact, the measured r0 of the ground atmosphere in Shanghai is between 2 cm and 16 cm, while the typical values for a satellite-to-ground link are 10 cm on average and up to 20 cm under excellent seeing conditions for a typical astronomical optical observatory [35]. Therefore, the noise level of the 16 km atmosphere link is greater than that of the GEO link. The average loss for the atmospheric link was approximately 52 dB.

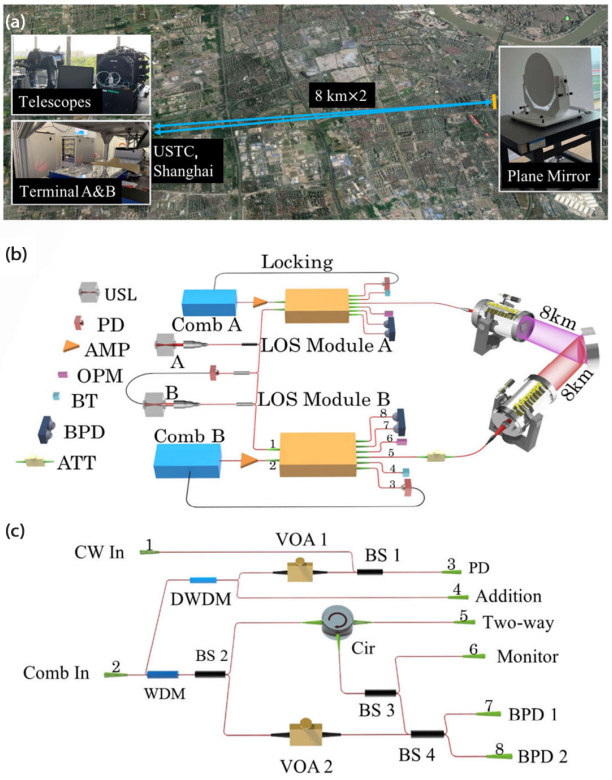


Fig. 3. (a) Transfer takes place at Pudong, Shanghai, between two co-located sites, terminals A and B, with synchronized reference clock sources. The sites are linked by two 70 m optical fiber paths from the laboratory to each free-space launch for the 16 km air path. (b) Our experimental setup. USL, ultrastable laser; PD, photodiode; AMP, amplifier; BT, beam trap; ATT, attenuator; OPM, optical power meter; BPD, balanced photon diode; and LOS module, a compact box for linear optical sampling. (c) Detailed structure of the LOS module. WDM, a 20 nm bandpass filter centered at 1,520 nm; DWDM, a 0.8 nm narrow filter centered at 1,550.12 nm; Cir, circulator; VOA 1 and VOA 2, attenuators; BS 1, 50:50 beam splitter for a beat between the CW laser and comb; BS 2, 99:1 beam splitter, and its 99% port is connected to the circulator; BS 3, 95:5 beam splitter, and its 5% port is connected to the monitor port; BS 4, 50:50 beam splitter for a beat between the local comb and the signal.

Figure 3(b) shows an illustration of the experiment setup. Both terminals A and B are equipped with an ultrastable laser as the reference clock source. One laser was phase-locked to the other, also enabling time synchronization and system calibration. The comb on each side was phase-locked to the corresponding ultrastable laser with in-loop relative frequency instabilities of $1 - 2 \times 10^{-16}$ at 1 s and 2×10^{-22} at 10,000 s, evaluated with a modified Allan deviation. Repetition rates of the combs were set with a difference of $\Delta f_r \approx 2.6$ kHz to produce periodic optical interferograms. One interferogram is named as an interference frame or one sample of time measurement. The duration between adjacent frames is approximately 380 μs. The heterodyne interference signal from the balanced detector is sampled by a commercial acquisition card with two-channel, 14-bit analog-to-digital converters (ADCs). The card is operated at the threshold trigger mode. Only the interference frames with a maximum signal amplitude higher than the trigger threshold are saved for further process. In our experiment, to tolerate a loss as large as possible, the hardware trigger threshold of the acquisition card is set at a considerably low level. This also introduces additional noise that must be removed by post-processing. (Supplement 1 has a detailed description of the

noise-removing method.) The Δf_r is a key parameter for LOS. According to a previous study [36], a low Δf_r often induces high precision and can help improve detection sensitivity. A larger Δf_r decreases the energy of each interferogram and causes a poor temporal signal-to-noise ratio (SNR). This increases the difficulty of selecting a proper trigger threshold. The Doppler effect provides a lower limit for the selected Δf_r as the satellite moves closer or farther away from the ground station. The actual Δf_r when the signal reaches the ground station varies in some range. However, if the Δf_r is zero, the LOS would stop working. For the repetition frequency of 250 MHz of OFCs in our setup, the frequency offset due to the Doppler effect is up to 2.5 kHz, corresponding to a 3 km/s speed, as shown in Fig. 2, for a typical MEO orbit. The optical signal of the OFCs was filtered by a 20 nm bandpass filter centered at 1520 nm, ensuring that the spectrum width of a beat note between OFCs is below the Nyquist frequency of LOS so that the LOS does not produce any aliasing. The filtered laser was sent to telescopes via a 70 m standard single-mode fiber for each end, where a nonzero dispersion-shifted fiber 100 m in length with negative dispersion of -4.0 ps/nm · km was added at one end to compensate for the entire dispersion of the link. To simulate the loss of a GEO link, we inserted an adjustable attenuator and set the average loss up to 72 dB.

To overcome such a large loss, we maximized the optical coupling efficiency via polarization manipulation and employed high-power OFCs. As shown in Fig. 3(c), the additional loss of the LOS module with a circulator directing laser signals in principle can be below 1 dB, which is more than 5 dB lower than the beam splitter approach [37]. This setup results in a slightly longer nonreciprocal path, degrading the symmetry of the link. Therefore, we deliberately matched the length of the optical paths and packed all parts in a small box with temperature stabilization; the noise floor was checked to be 10^{-19} level for a 10,000 s integration time. First, we use commercial amplifiers to increase the optical power from 5 mW to 500 mW. Unfortunately, only 120 mW remains after the 20 nm bandpass filter. The experiment of the 52 dB link and 64 dB link is performed with these amplifiers. To tolerate a higher channel loss, we built two homemade amplifiers with a 250 mW output power within 20 nm centered at 1540 nm. The duration of the laser pulse sent to the telescope was approximately 4 ps. The dispersion is not fully compensated, resulting in chirped pulses at the interference point, degrading the sensitivity of the LOS. Nonetheless, in real satellite–ground links, it is difficult for the atmospheric dispersion to be real-time compensated, thereby limiting the minimum pulse width to 1–2 ps for a 2 THz bandwidth comb [38]. Therefore, the 4 ps pulse duration in our experiment is sufficient to mimic the dispersion conditions for real satellite–ground time transfer. The relative intensity noise (RIN) of the amplified laser, which is approximately 10 dB higher than that of the seed laser, is approximately -80 dBc/Hz at 1 Hz and -130 dBc/Hz for frequencies higher than 100 kHz, which is approximately eight orders below the RIN induced by the air turbulence link.

4. TEST RESULTS

Figure 4(a) shows the received power of the 72 dB link, exhibiting a mean value of 10 nW measured at the monitoring port. As shown in the figure, the received power is extremely unstable, leading to failing samplings. To efficiently produce a value for a one-way time-of-flight (TOF), we preselected raw data from each single

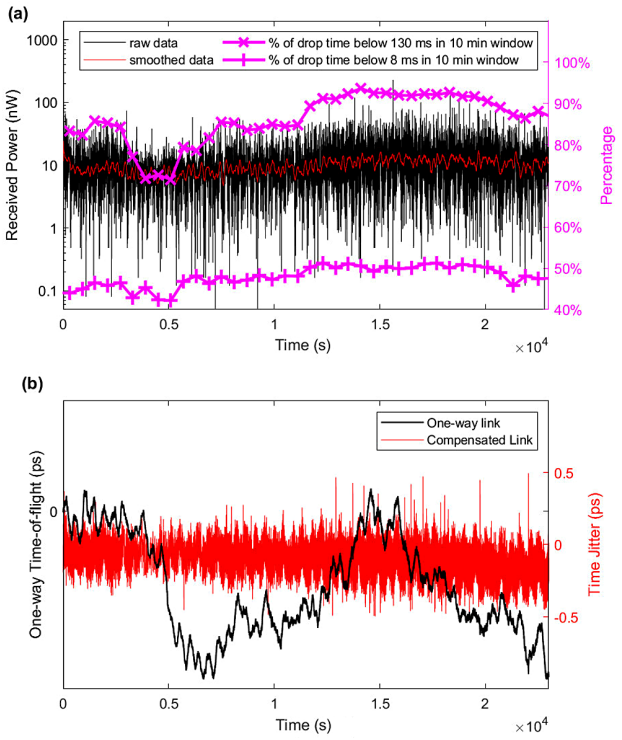


Fig. 4. (a) Received power of 72 dB link (one-second per point). The black line is the raw data, and the red line is the moving average-smoothed data with an average length of 100 s. The average loss over the entire duration was 72 dB. Proportions of dropout time below 130 ms and 8 ms are shown in purple (10 minutes per point). The dropout time is calculated by making the difference between the time tag of the adjacent data. (b). Time fluctuations of the 72 dB link. Fluctuation of the one-way TOF in black and the compensated link in red. The data were obtained from each overlapping process between two OFCs with raw data from a single frame.

frame by setting a threshold and calculated the TOF only if the maximum value of a single frame is beyond the threshold. With an appropriate threshold value, TOFs with large divergence can be avoided. The time jitter of the compensated link is approximately 60 fs [see Fig. 4(b)], while the time fluctuation of the one-way link to be 17 ps peak-to-peak in 20,000 s. The statistics of signal fade durations is useful to judge the link quality; therefore, we performed a statistical analysis of the dropouts to be 90% below 130 ms and 50% below 8 ms, as shown in Fig. 4(a). All data in Fig. 4 are acquired simultaneously.

Figure 5(a) shows the calculated time deviation (TDEV) of the link with different losses, and the system floor. The TDEV of the system floor reaches 0.3 fs at 10,000 s and is attributed to thermal effects owing to the asymmetric optical paths of the local system. We have checked the noise floor by decreasing both the optical power and attenuation, and no difference was observed. This indicates that the nonlinear effect due to the optical amplification process is negligible. For a medium link loss of 64 dB and 52 dB, the TDEV is almost below 10 fs between 1 s and 3,000 s, while it is a few 10 fs for the 72 dB link. The fractional frequency instabilities of the 72 dB link, as shown in Fig. 5(b), are 3×10^{-14} at 1s and 4×10^{-18} at 3,000 s. The bumps at tens of seconds are attributed to the air conditioner, which is not consistent on a daily basis owing to the weather involved with poor thermal isolation, leading to differences in these results. For short terms ($\tau < 10$ s), instability of the link is determined by the SNR and valid data rates. The

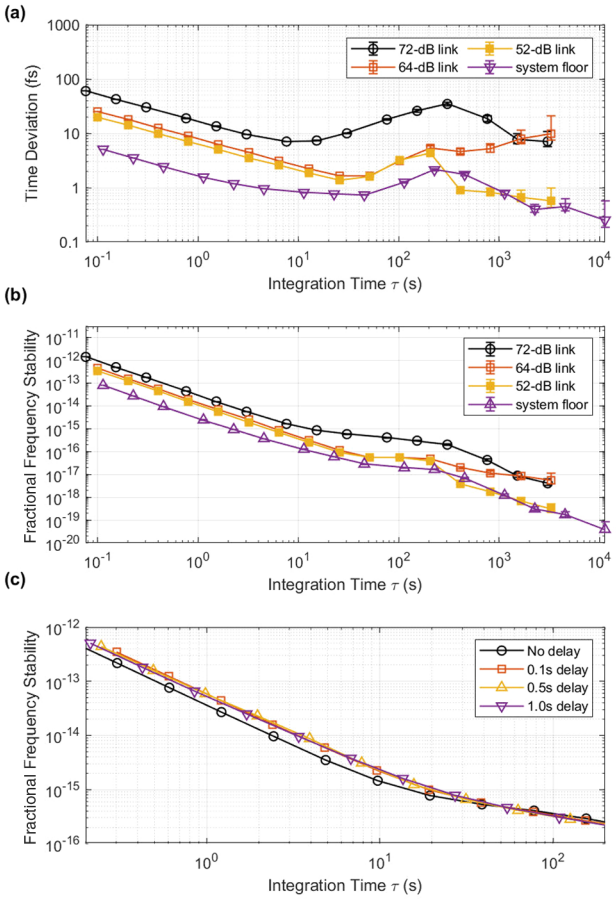


Fig. 5. (a) TDEV of the time transfer with different link losses, where the system floor is measured without the link. Large jitters due to atmospheric scintillation have been removed, and 91% of the raw data is used for the 72 dB link. (b) Modified Allan deviation of frequency transfer. (c) Results of asymmetric delay simulation with different delay times. These curves were calculated using the same data as the 72 dB link. The delay we added here mimics that of the satellite–ground link. A 0.1 s delay, for example, corresponds to a 30,000 km height orbit. A 1 s delay is even close to a link delay between the Earth and the moon.

received signal power influences the SNR and the single-frame timing precision of LOS [36], especially when the received power is close to the sensitivity power threshold. The timing precision of a single frame is 25 fs and 20 fs for the 64 dB and 52 dB links, respectively, corresponding to the average received signal power of 28 nW and 440 nW. However, for the 72 dB link, the timing precision degrades to 60 fs owing to a poor SNR. When the SNR is sufficiently high, the instability is dominated by data rates. Unfortunately, the data transmission speed from the acquisition card to the computer is limited to approximately 10 frames per second (fps) for medium-loss links. Actually, we expect to gain a factor of approximately 16 (i.e., $\sqrt{2600/10}$) for low-loss links with a high-speed data acquisition system because the repetition rate difference between the two OFCs is 2.6 kHz.

The satellite–ground link has a longer delay and worse symmetry owing to the atmosphere located on one side. To evaluate the instability of a real link, we perform a simulation by shifting the time tag of one side by Δt with the raw data of the 72 dB link. In the original two-way time transfer, we combined the time series of both sides and extracted data with the same arrival time. Here, we shift the arrival times of the time series on one side to obtain a new time series and subsequently extract data using the same process. If we assume that no additional link noise exists out of the atmosphere, the time-shifted data should be equivalent to a long link combined by a 16 km air path and $c \times \Delta t$ length out of the atmosphere. Figure 5(c) shows the simulation results of satellite–ground links up to 300,000 km; the deterioration of performance is apparent in the short term and disappears after 40 s. The noise effect owing to a long delay and asymmetric noise distribution of a GEO link should not be a long-term limitation at the level we obtained. In addition, the instabilities of the time-shifted link are almost the same, implying that the noise contribution of the free-space link is mainly located in the range of tens to thousands of hertz. We can further investigate this by using a high-speed data acquisition system.

5. CONCLUSION

In summary, we analyzed the various effects of different satellite orbits with regard to time–frequency transfer. Despite the challenges of a large link loss and long transmission delay, GEO is the optimal choice for intercontinental optical clock comparison while considering relativistic effects, Doppler effects, and passing and common view time. We have demonstrated the optical frequency transfer through a 16 km free space to simulate the loss, noise, and delay effect of a GEO link, and we obtained an instability of 3×10^{-14} at 1 s and 4×10^{-18} at 3,000 s. High channel loss was overcome by increasing the optical power and improving the coupling efficiency as well as tolerating a low received power of 10 nW. Based on these results, we expect that a 10^{-18} GEO link might be realized at 10,000 s, and the fundamental limitations are still far from what we performed. We believe such a link can not only provide remote optical atomic clock comparison [26] but also find immediate applications such as quantum communications, including free-space twin-field quantum key distribution [39] and quantum repeaters [40].

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Supplemental document. See Supplement 1 for supporting content.

[†]These authors contributed equally to this work.

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